

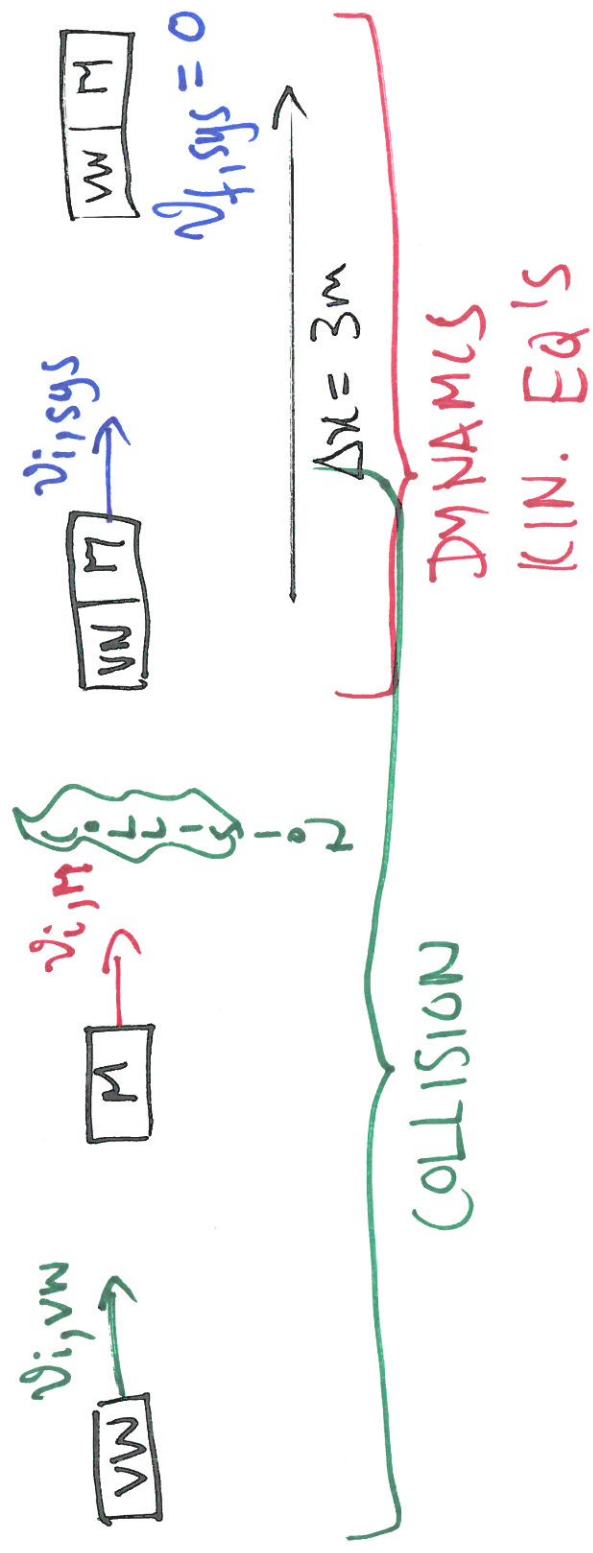
Example

A 2000kg Mercedes just started moving ~~for~~ forward from a stop sign when it was struck from behind by a 1000kg Volkswagen.

The bumpers of the cars became entangled during the collision and the two cars skidded forward until they came to rest 3m from the stop sign. The driver of the Mercedes gave testimony that she was moving at a speed of 5m/s just before the collision.

Did the Volkswagen exceed the 60 km/h speed limit?

Assume friction between the tyres and the road with $\mu_k = 0.8$.



For the collision: Cons. of lin. mom.

Prat, i = p tot, f

$$m_{VW} v_{i,VW} + m_M v_{i,M} = (m_{VW} + m_M) v_{i,sys}$$

$$v_{i,VW} = \frac{(m_{VW} + m_M) v_{i,sys} - m_M v_{i,M}}{m_{VW}}$$

$$= \frac{(3000\text{kg})(6.9\text{m/s}) - (2000\text{kg})(5\text{m/s})}{1000\text{kg}}$$

from dynamics calculation

$$v_{i,VW} = 10.7\text{m/s} \approx 38.5\text{km/h}$$

Dynamics



$$\Sigma F_y = 0$$

$$N_{vw+m} = (m_{vw} + m_m)g$$

$$\text{But } f_k = \mu_k (m_{vw} + m_m)g \quad - (1)$$

$$\Sigma F_x = - (m_{vw} + m_m) a_x$$

$$-f_k = - (m_{vw} + m_m) a_x$$

$$a_x = \frac{f_k}{m_{vw} + m_m} = \frac{\mu_k (m_{vw} + m_m)g}{(m_{vw} + m_m)}$$

$$a_x = \mu_k g = (0.8)(9.8 \text{ m/s}^2)$$

$$a_x = 7.84 \text{ m/s}^2$$

$$v_{f, \text{sys}}^2 = v_{i, \text{sys}}^2 - 2 a_x \Delta x$$

$$v_{i, \text{sys}} = 2 a_x \Delta x = 2(7.84 \text{ m/s}^2)(3 \text{ m}) = \dots$$

$$v_{i, \text{sys}} = 6.9 \text{ m/s}$$